

Cement and Concrete Terminology

Reported by ACI Committee 116

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FOREWORD

This report is the authoritative glossary for cement and concrete technology. It is to be used generally and specifically in ACI technical communications, correspondence, and publications. One mission of Committee 116 is to produce and maintain a list of terms with their meaning in the field of cement and concrete technology.

Committee 116 has tried to produce a glossary that will be useful, comprehensive, and up-to-date. It recognizes, however, that the listing may not be complete and that some definitions may be at variance with some commonly accepted meanings.

Users of the glossary are invited to submit suggestions for changes and additions to ACI Headquarters for consideration by Committee 116 when preparing future editions. In the event that a user disagrees with any of the definitions, it is hoped that the reasons for such will be given to the committee.

The committee is aware that some of the definitions included may seem entirely self-evident to an expert in the concrete field. This occurs because no term has been discarded if there was reason to believe it would appear to be technical in nature to a casual reader of the ACI literature.

The committee voted to use the following procedural rules:

1. Each definition shall be stated in one sentence;
2. Each definition shall consist of the term printed in boldface, a dash, and the definition statement;
3. The definition statement shall not repeat the term and should state the class or group and identify the features unique to the term; as “**mathematics**—the science of numbers and spaces”;
4. Verbs should be stated in the infinitive rather than the participle; for example the term to be defined should be “**abrade**” not “**abrading**”;
5. Notes may be appended to definition statements;
6. Cross references may take the place of a definition as “**green concrete**—see **concrete, green**.” They also may call attention to related items as “**flint**—a variety of chert. (See also **chert**).” Where the committee has found two or more terms with the same meaning, the definition is given where the preferred term appears, the synonyms are cross referenced to the preferred term, and in many cases, the fact is stated;
7. Generally, where there are a number of terms, the last word of which is the same, the definitions are given where the terms are listed in the inverted form, as “**cement, low-heat**” rather than “**low-heat cement**,” but under the latter entry, there will be a cross reference “see **cement, low-heat**,” and
8. In selecting terms and definitions, there shall be coordination with the terminology subcommittees of ASTM Committees C-1 on Cement, and C-9 on Concrete and Concrete Aggregates.

The invaluable contributions of the past chairmen of Committee 116, B. J. Erlin, R. C. Mielenz, D. L. Bloem, W. H. Price, R. E. Davis, Jr., J. R. Dise, K. F. Gibbe, Robert L. Henry, M. D. Luther, B. Mather, and E. Senbetta, those of the present members of the committee, as well as the diligent efforts of William Lorman and Lewis H. Tuthill, are gratefully acknowledged.

For drafting this edition, all members, both associates and voting, participated.

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Reference to this document shall not be made in contract documents. If items found in this document are desired by the Architect/Engineer to be a part of the contract documents, they shall be restated in mandatory language for incorporation by the Architect/Engineer.

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A

Abrams' law—see **law, Abrams'**.

abrasion damage—see **damage, abrasion**.

abrasion resistance—see **resistance, abrasion**.

absolute specific gravity—see **specific gravity, absolute**.

absolute volume—see **volume, absolute**.

absorbed moisture—see **moisture, absorbed**.

absorbed water—see **moisture, absorbed**.

absorption—the process by which a liquid is drawn into and tends to fill permeable voids in a porous solid body; also, the increase in mass of a porous solid body resulting from the penetration of a liquid into its permeable voids.

abutment—in bridges, the end structure (usually of concrete) that supports the beams, girders, and deck of the bridge, or combinations thereof, and sometimes retains the earthen bank or supports the end of the approach pavement slab; in prestressing, the structure against which the tendons are stressed in producing pretensioned precast members or post-tensioned pavement; and in dams, the side of the gorge or bank of the stream against which a dam abuts.

accelerating admixture—see **admixture, accelerating**.

acceleration—increase in velocity or in rate of change, especially the quickening of the natural progress of a process such as setting or strength development (hardening) of concrete. (See also **admixture, accelerating**.)

accelerator—see **admixture, accelerating**.

accidental air—see **air, entrapped**.

acrylic resin—see **resin, acrylic**.

addition—a material that is interground or blended in limited amounts into a hydraulic cement during manufacture either as a "processing addition" to aid in manufacturing and handling the cement or as a "functional addition" to modify the use properties of the finished product.

advancing-slope grouting—see **grouting, advancing-slope**.

additive—see **agent**.

adhesion—the state in which two surfaces are held together by interfacial effects that may consist of molecular forces, interlocking action, or both.

adhesives—the group of materials used to join or bond similar or dissimilar materials; for example, in concrete work, the epoxy resins.

adiabatic—a condition in which heat neither enters nor leaves a system.

adiabatic curing—see **curing, adiabatic**.

adjustment screw—see **screw, adjustment**.

admixture—a material other than water, aggregates, hydraulic cement, and fiber reinforcement, used as an ingredient of a cementitious mixture to modify its freshly mixed, setting, or hardened properties and that is added to the batch before or during its mixing.

admixture, accelerating—an admixture that causes an increase in the rate of hydration of the hydraulic cement and thus shortens the time of setting, increases the rate of strength development, or both.

admixture, air-entraining—an admixture that causes the development of a system of microscopic air bubbles in concrete, mortar, or cement paste during mixing, usually to increase its workability and resistance to freezing and thawing. (See also **air, entrained**.)

admixture, retarding—an admixture that causes a decrease in the rate of hydration of the hydraulic cement and lengthens the time of setting.

admixture, water-reducing—an admixture that either increases slump of freshly mixed mortar or concrete without increasing water content or maintains slump with a reduced amount of water, the effect being due to factors other than air entrainment.

admixture, water-reducing (high-range)—a water-reducing admixture capable of producing large water reduction or great flowability without causing undue set retardation or entrainment of air in mortar or concrete.

adobe—unburnt brick dried in the sun.

adsorbed water—see **water, adsorbed**.

adsorption—development (at the surface of either a liquid or solid) of a higher concentration of a substance than exists in the bulk of the medium; especially formation of one or more layers of molecules of gases, of dissolved substances, or of liquids at the surface of a solid (such as cement, cement paste, or aggregates), or of air-entraining agents at the air-water interfaces; also, the process by which a substance is adsorbed. (See also **water, adsorbed**.)

advancing-slope grouting—see **grouting, advancing-slope**.

advancing-slope method—see **method, advancing-slope**.

aerated concrete—see **concrete, cellular** and **concrete, foamed**.

A/F ratio—see **ratio, A/F**.

afwillite—a mineral with composition $3\text{CaO} \cdot 2\text{SiO}_2 \cdot 3\text{H}_2\text{O}$ occurring naturally in South Africa, Northern Ireland, and California, and artificially in some hydrated portland cement mixtures.

agent—a general term for a material that may be used either as an addition to cement or an admixture in concrete; for example, an air-entraining agent.

agent, air-entraining—see **admixture, air-entraining**.

agent, bonding—a substance applied to a suitable substrate to create a bond between it and a succeeding layer.

agent, parting—see **agent, release** (preferred term).

agent, release—material used to prevent bonding of concrete to a surface. (See also **bond breaker** and **oil, form**.)

agent, surface-active—a substance that markedly affects the interfacial or surface tension of solutions when present even in low concentrations.

agent, wetting—a substance capable of lowering the surface tension of liquids, facilitating the wetting of solid surfaces, and permitting the penetration of liquid into the capillaries.

agglomeration—a gathering into a ball or mass.

aggregate—granular material, such as sand, gravel, crushed stone, crushed hydraulic-cement concrete, or iron blast-

furnace slag, used with a hydraulic cementing medium to produce either concrete or mortar. (See also **aggregate, heavyweight** and **aggregate, lightweight**.)

aggregate, angular—aggregate particles that possess well-defined edges formed at the intersection of roughly planar faces.

aggregate, coarse—aggregate predominantly retained on the 4.75 mm (No. 4) sieve, or that portion retained on the 4.75 mm (No. 4) sieve. (See also **aggregate**.)

aggregate, crusher-run—aggregate that has been mechanically broken and has not been subjected to subsequent screening.

aggregate, dense-graded—aggregates graded to produce low void content and maximum density when compacted. (See also **aggregate, well-graded**.)

aggregate, fine—aggregate passing the 9.5 mm (3/8 in.) sieve, almost entirely passing the 4.75 mm (No. 4) sieve, and predominantly retained on the 75 μ m (No. 200) sieve; or that portion passing the 4.75 mm (No. 4) sieve and predominantly retained on the 75 μ m (No. 200) sieve. (See also **aggregate** and **sand**.)

aggregate, gap-graded—aggregate graded so that certain intermediate sizes are substantially absent.

aggregate, heavyweight—aggregate of high density, such as barite, magnetite, hematite, limonite, ilmenite, iron, or steel, used in heavyweight concrete.

aggregate, lightweight—aggregate of low density, such as: a) expanded or sintered clay, shale, slate, diatomaceous shale, perlite, vermiculite, or slag; b) natural pumice, scoria, volcanic cinders, tuff, and diatomite; and c) sintered fly ash or industrial cinders, used in lightweight concrete.

aggregate, mineral—aggregate consisting essentially of inorganic nonmetallic rock materials, either natural or crushed and graded.

aggregate, normalweight—aggregate that is neither heavyweight nor lightweight.

aggregate, open-graded—aggregate in which the voids are relatively large when the aggregate is compacted.

aggregate, reactive—aggregate containing substances capable of reacting chemically with the products of solution or hydration of the portland cement in concrete or mortar under ordinary conditions of exposure, resulting in some cases in harmful expansion, cracking, or staining.

aggregate, refractory—aggregate having refractory properties that, when bound together into a conglomerate mass by a matrix, forms a refractory body.

aggregate, single-sized—aggregate in which a major portion of the particles is in a narrow size range.

aggregate, well-graded—aggregate having a particle-size distribution that produces maximum density, that is, minimum void space.

aggregate blending—the process of intermixing two or more aggregates to produce a different set of properties, generally, but not exclusively, to improve grading.

aggregate-cement ratio—see **ratio, aggregate-cement**.

aggregate gradation—see **grading** (preferred term).

aggregate interlock—the effect of portions of aggregate particles from one side of a joint or crack in concrete protruding into recesses in the other side of the joint or crack so as to transfer load in shear and maintain alignment.

aggregate transparency—discoloration of a concrete surface consisting of darkened areas over coarse aggregate particles immediately below the concrete surface.

agitating speed—see **speed, agitating**.

agitating truck—see **truck, agitating**.

agitation—

1. the process of providing motion in mixed concrete just sufficient to prevent segregation or loss of plasticity; and
2. the mixing and homogenization of slurries or finely ground powders by either mechanical means or injection of air. (See also **agitator**.)

agitator—a device for maintaining plasticity and preventing segregation of mixed concrete by agitation. (See also **agitation**.)

aids, grinding—materials used to expedite the process of grinding by eliminating ball coating, dispersing the finely ground product, or both.

air—

air, accidental—see **air, entrapped** (preferred term).

air, entrained—microscopic air bubbles intentionally incorporated in mortar or concrete during mixing, usually by use of a surface-active agent; typically between 10 and 1000 μ m (1 mm) in diameter and spherical or nearly so. (See also **air entrainment**.)

air, entrapped—air voids in concrete that are not purposely entrained and that are larger, mainly irregular in shape, and less useful than those of entrained air; and 1 mm or larger in size.

air blow pipe—air jet used in shotcrete gunning to remove rebound or other loose material from the work area.

air-blown mortar—see **shotcrete** (preferred term).

air content—the volume of air voids in cement paste, mortar, or concrete, exclusive of pore space in aggregate particles; usually expressed as a percentage of total volume of the paste, mortar, or concrete.

air-cooled blast-furnace slag—see **blast-furnace slag**.

air entraining—the capability of a material or process to develop a system of microscopic bubbles of air in cement paste, mortar, or concrete during mixing. (See also **air entrainment**.)

air-entraining agent—see **admixture, air-entraining**.

air-entraining hydraulic cement—see **cement, air-entraining hydraulic**.

air entrainment—the incorporation of air in the form of microscopic bubbles (typically smaller than 1 mm) during the mixing of either concrete or mortar. (See also **air entraining** and **air, entrained**.)

air lift—equipment whereby slurry or dry powder is lifted through pipes by means of compressed air.

air meter—see **meter, air**.

air-permeability test—see **test, air-permeability** and **test, Blaine**.

air ring—see **ring, air**.